ASTRA polytechnic
member of ASTRA

Kampus Merdeka INDONESIA JAYA
satu Indonesia

Introduction to Algorithm

Prepared by : Radix R

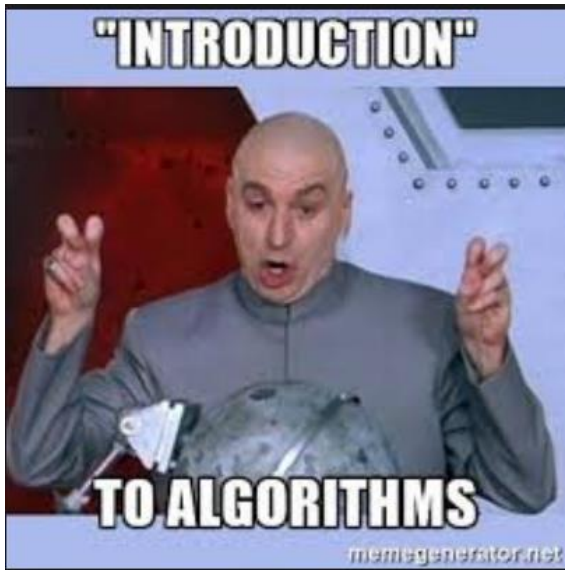
Contents

- Intro to Algorithm
- Algorithm Characteristic
- Program and Programming
- Programming Language
- Algorithm Notation
- Procedural Programming
- Instruction and Action
- 3 Basic Construction
- Pseudo-Code Notation
- Algorithm Text Structure

Programming 1

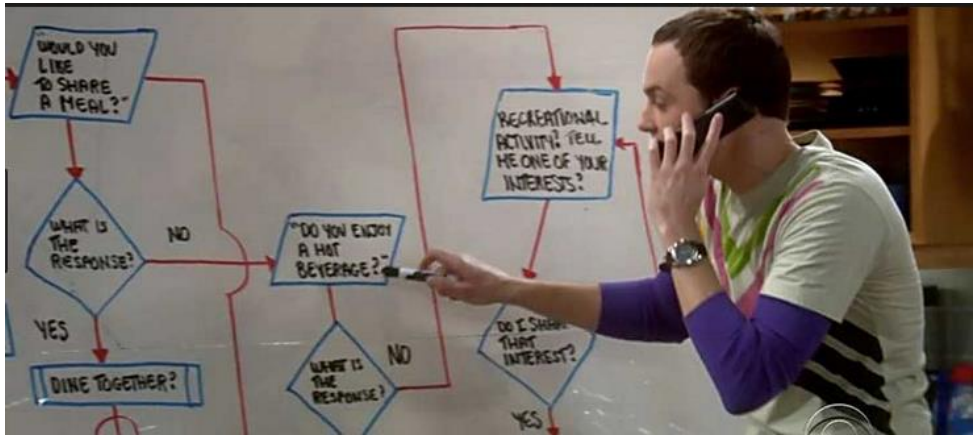
- 4 Credits
- 3 days ~ 12 hours/wk
- Practice 60%, Theory 40%
- Your Score :
  - Exam : Mid Exam & Final Exam
  - Quiz & Practice deliverables
  - Final Project

Intro To Algorithm



What is Algorithm ?





Algorithm

- What exactly is an algorithm? Algorithms explained | BBC Ideas

https://www.youtube.com/watch?v=ZnBF2GeAKbo

- What's an algorithm? - David J. Malan

https://www.youtube.com/watch?v=6hfOvs8pY1k

Anders Dilemma



Anders Andert Anderson left for a place in Alaska that took six days to walk from Shungnak across a deserted snow and ice. One person can only bring enough food and water for a four-day supply. So, one person cannot go alone because his supply will be exhausted before he reaches his destination. How many people are needed to carry the supplies so that Anders can clearly reach the destination and his assistants can return to Shungnak?

hints:

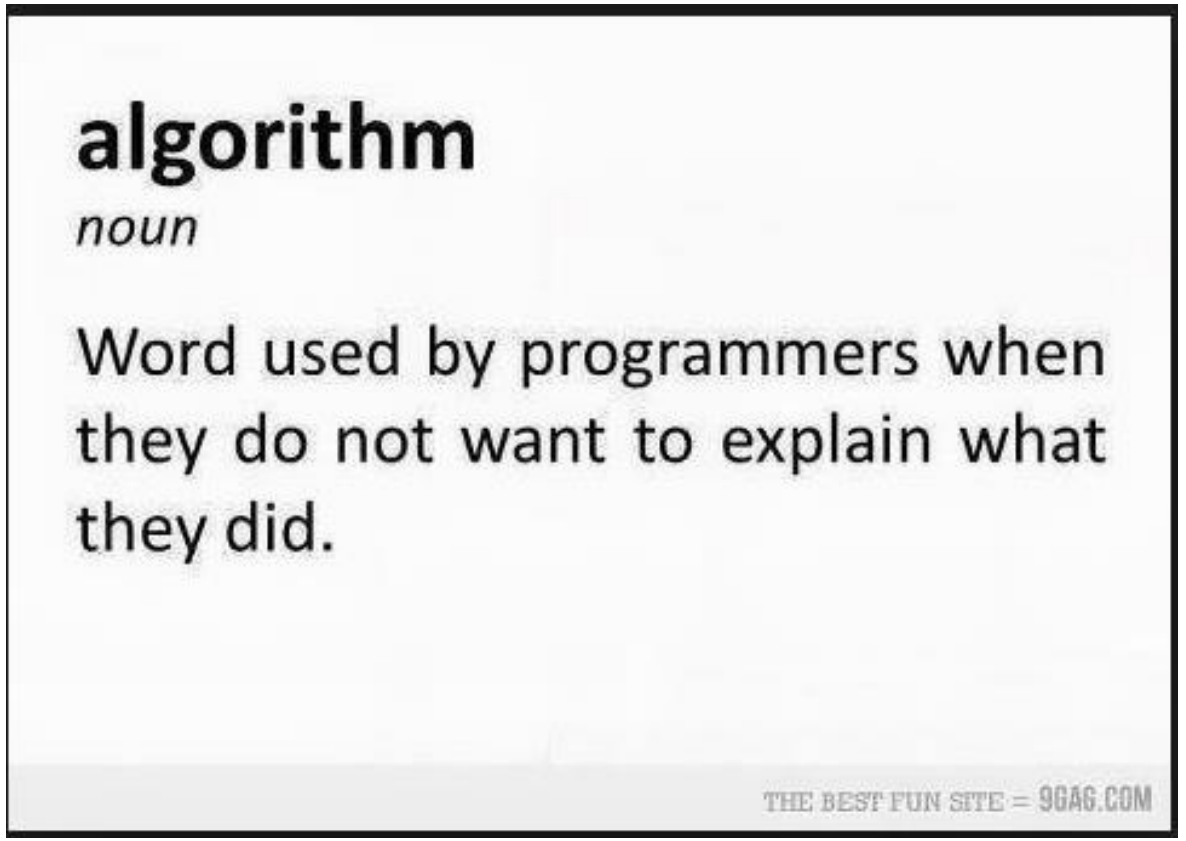
No matter how many people left for Shungnak, only Anders had to get to his destination. But remember; Assistants also need to eat, so make sure that others get enough food so they can return home.

Anders Dilemma

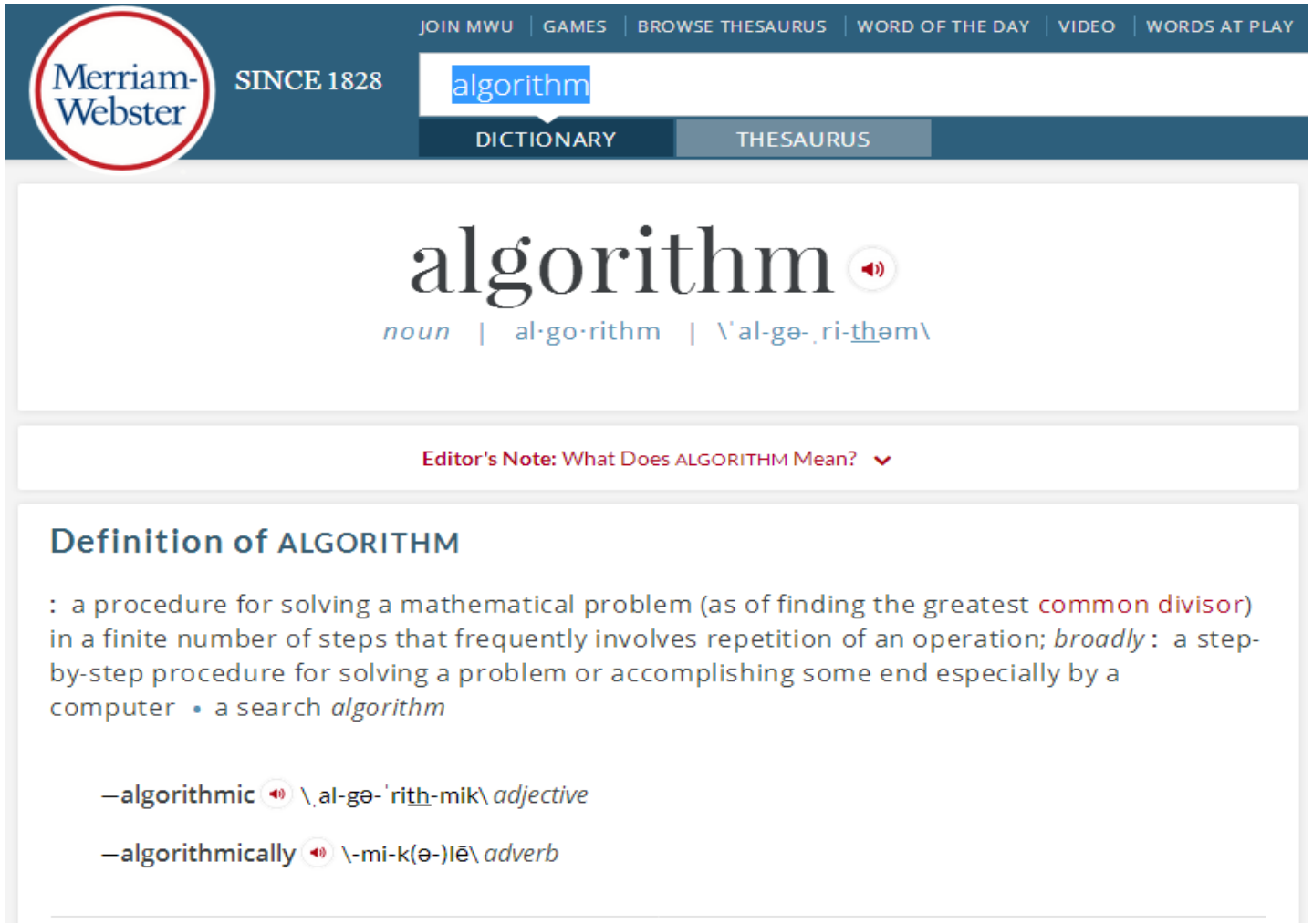


It takes **three** people. One assistant brought supplies for four days and after the first day's trip, he gave one day's supplies to another assistant and another to Anders. this makes them have full inventory for four days. The first assistant, with the remaining inventory a day for himself, returns home. The other two people finished the trip the next day and after the day ended, the assistant gave Anders a day's supply, which again had a four-day supply, enough to get to the destination. His assistant still has enough supplies to get back to his house in two days.

Intro To Algorithm



Intro To Algorithm



Intro To Algorithm



Problems

Merriam-Webster

1 problem

noun | prob·lem | \ˈprä-bləm, -bᵊm, -ˌblem\

1 **a :** a question raised for inquiry, consideration, or solution
b : a proposition in mathematics or physics stating something to be done



- Given a deck of cards. Sort the cards so they are arranged ascending by their numbers such that the small numbers are above and the big numbers are below!

Problems

1. Given a deck of cards. Sort the cards so they are arranged ascending by their numbers such that the small numbers are above and the big numbers are below!
2. Given a table of test participants who passed the test. Are there any participant with test number X in the list?
3. Given a list of cities in the world with current air temperatures. Determine the city with the highest temperature!
4. Given a rectangular spot of land with a certain length and width (in meters). Calculate the area of the land!

Generic Problems

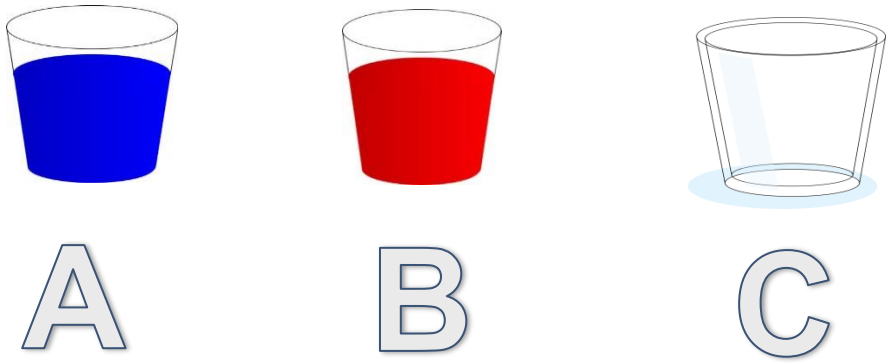
1. [Sorting problems] Given a S list containing n values. Sort S so it's arranged in ascending order!
2. [Searching problems] Given a S list containing n values. Is an x value contained in S?
3. [Searching for maximum value problems] Given a S list containing n values. Find the maximum value in S!
4. [Calculate rectangular area problems] Given a rectangular with length a and width b (in meters). Calculate the area of the rectangular!

Definition of an Algorithm

Algorithm is …

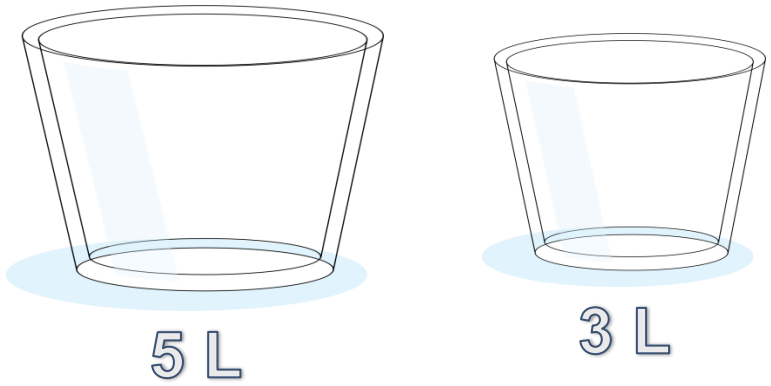
a sequence of steps to solve a problem.

Exchanging two glass content problem



- The algorithm to exchange the contents of two glasses A and B:
  1. Pour the solution from glass A to glass C
  2. Pour the solution from glass B to glass A
  3. Pour the solution from glass C to glass B

Water Jug Problem



- How to get 4L (Liters) of water from the lake using only this 2 jug of 5L and 3L?

More Problems & Algorithm

- Social Network – Hacking Scene
- Social Network – Algorithm Scene

- Social Network – Hacking Scene
- Social Network – Algorithm Scene

Algorithm Characteristic

- It must terminate after a finite number of steps
- There can be no ambiguity
- It takes zero or more inputs
- It results in zero or more outputs
- It must be effective

Program and programming

- Program is an algorithm written in computer language.
- Programming language is the computer language used to write program.
- Programming is an activity from designing to writing a program.
- Programmer is the person who wrote computer programs.

Program and programming

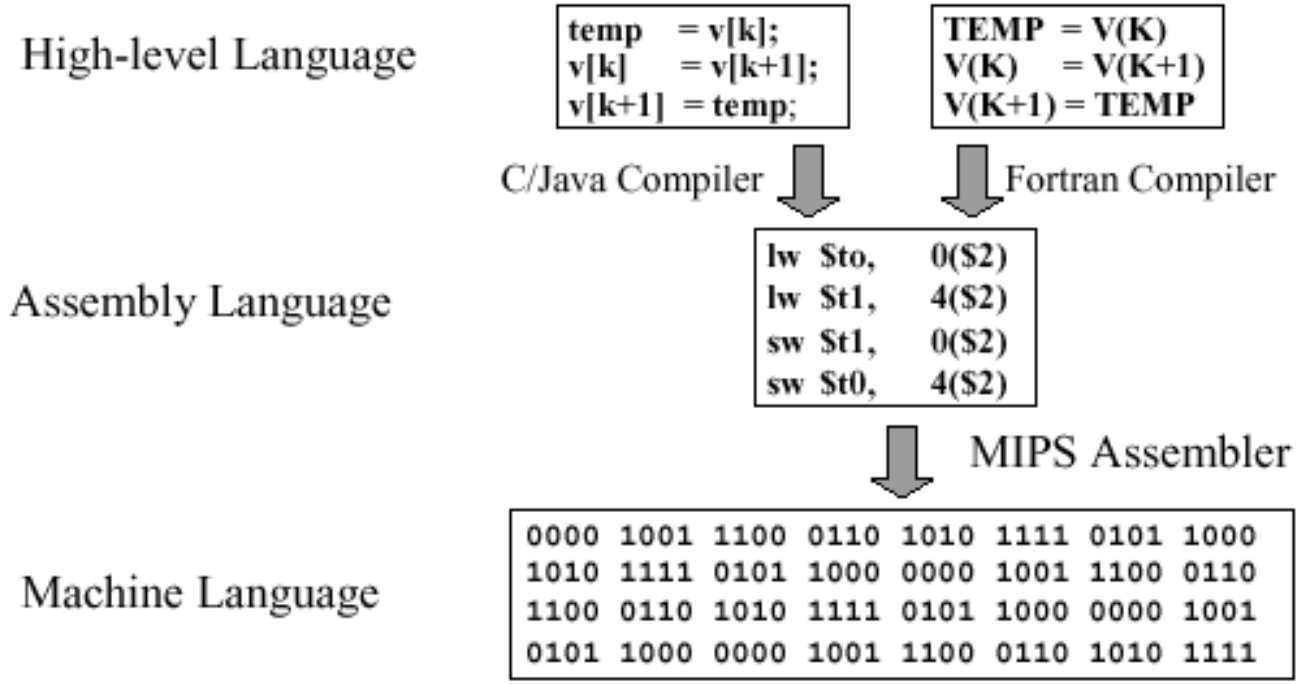
Learning programming ≠ Learning programming language

- Thinking about algorithm
- Thinking about design
- Independent on the programming language to be used

- Notation used
- Rules on each programming language

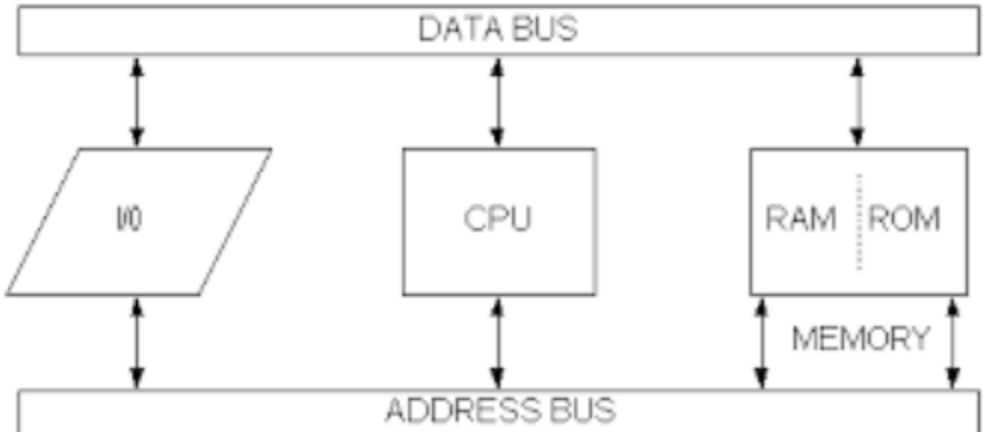
Programming Language

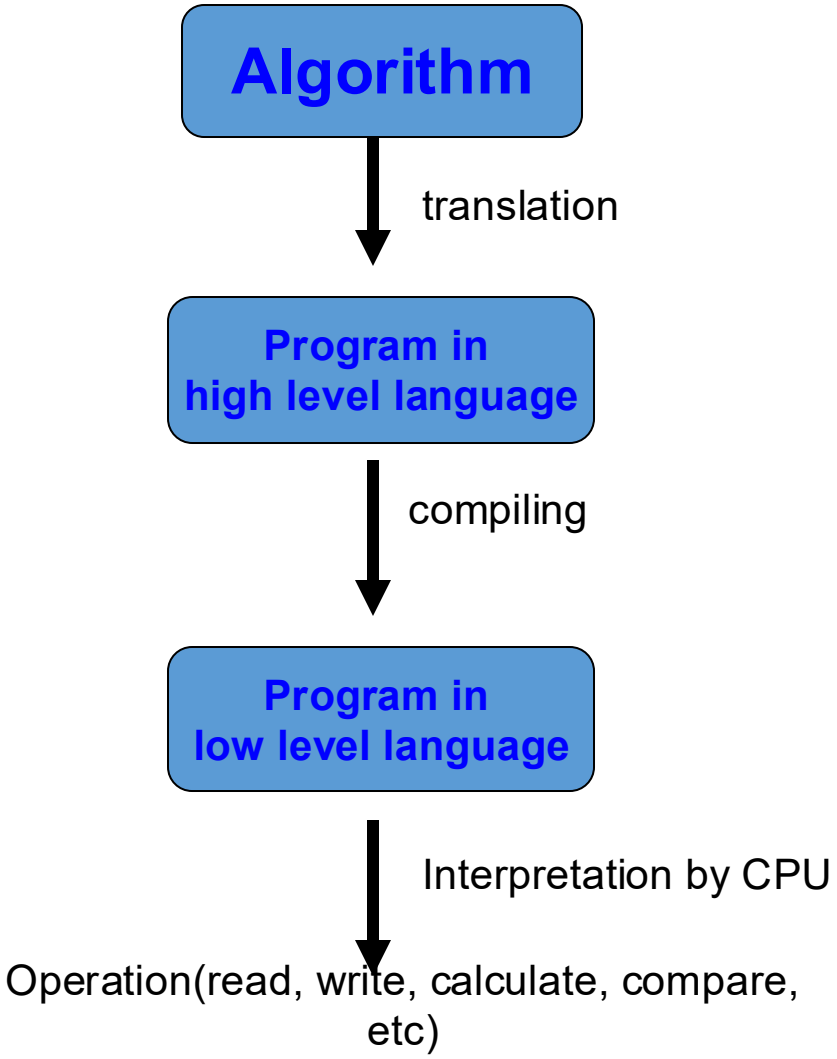
- Low level language : Assembly, Machine lang.
- High level language : Pascal, C, Java, Fortran, Basic, C#



Source : https://www.cise.ufl.edu/~mssz/CompOrg/CDA-lang.html

A program phase





Algorithm Notation

1. Descriptive Notation

- The algorithm is expressed by a descriptive sentence
- This notation is difficult to translate directly into the programming language notation

```
PROGRAM Euclidean

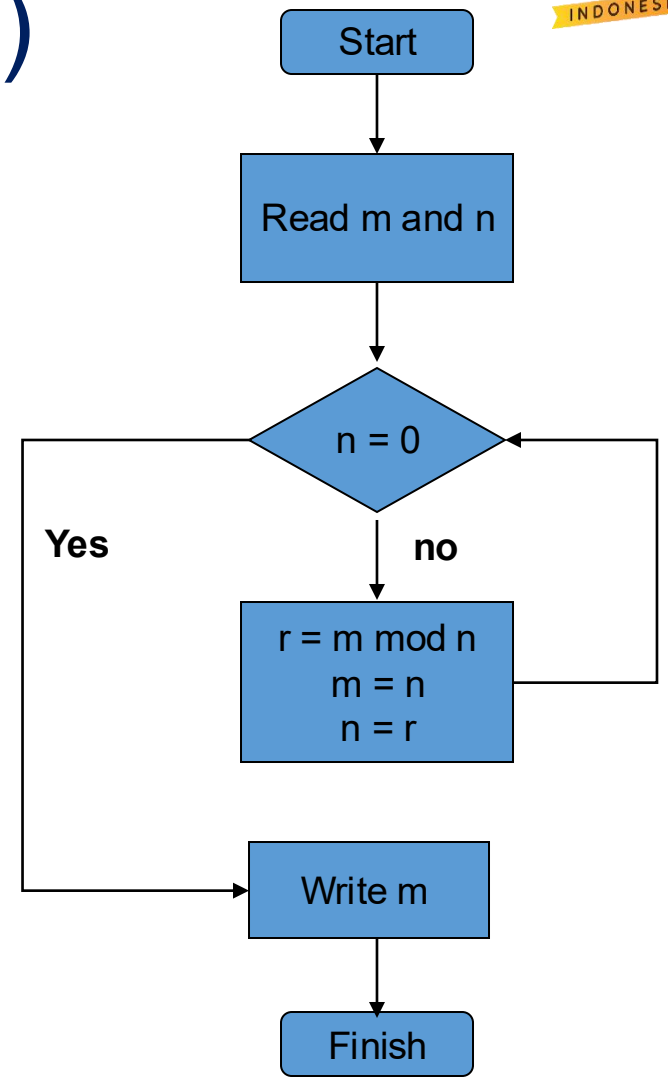
{Given two non-negative integers m and n (m ≥ n). The Euclidean algorithm
looks for the greatest common divisor, GCD, of the two numbers, the largest
number that divides both of them without leaving a remainder.}


ALGORITHM:
   1. If n = 0, then m is the answer. Stop.
      If n ≠ 0, go to step 2.
   2. Divide m by n, let r be the remainder.
   3. Replace the value of m with value of n, and value of n with value of r
   4. Return to step 1
```

Algorithm Notation (cont.)

2. Flowchart Notation

- Use certain symbols to describe the process
- Less practical to use for large and complex programs
- Relatively difficult to translate into programming language notation
- Example: from the example of the previous problem pictured here on the side.



Algorithm Notation (cont.)

3. Pseudo-Code Notation

- Using notation or a way that resembles high level programming language notation
- More easily translated into specific programming languages

```
PROGRAM Euclidean

{Given two non-negative integers m and n (m ≥ n). The Euclidean algorithm
looks for the greatest common divisor, GCD, of the two numbers, the largest
number that divides both of them without leaving a remainder.}

DECLARATION:
      m, n  : integer   { integers to be searched for their GCD }
      r     : integer   { remainder from division }

ALGORITHM:

Read (m, n)          { assume m ≥ n }
While n ≠ 0  do
      r  ← m mod n     { divide m with n, store the remainder in r }
      m ← n            { exchange m with n }
      n ← r            { exchange n with r }
End while
{ final condition after looping: n = 0 }

Write (m)
```

Procedural programming

- Procedural Programming is a programming that the steps of the program are executed sequentially or procedural.

- Basic, Pascal, Fortran, C, and C++ are procedural programming languages.

- Other programming paradigm are: Object Oriented Programming (OOP), Functional Programming, Declarative Programming.

Instruction and Action

- Algorithm contains a set of instructions that describes the problem-solving steps.
- Instructions are commands to perform a specific action.
- Example:
  - Write “Hello, World!”
  - Divide X with 2

3 Basic Construction

- **1. SEQUENCE**
- It consists of one or more statements
- Will be processed in accordance with the order of writing

- **2. SELECTION**
- It consists of one or more statements
- Will be processed if certain conditions are met

- **3. REPETITION**
- It consists of one or more statements
- Will repeat the process if certain conditions are met

SEQUENCE

Algorithm :

Statement 1

Statement 2

Statement 3

So statement 1 will be processed first, then statement 2 and then statement 3.

SELECTION

What if a statement in a sequence is processed when a condition is met?

Statement 1
Statement 2
Statement 3 → a condition
Statement 4
Statement 5

Statement 4 will be processed if Statement 3 is met

Statement 5 will be processed if Statement 3 is not met



Statements 4 and 5 are impossible to do both, only one will do.

SELECTION

Example :

Algorithm for watching adult movie theaters :

1. Go to the theater
2. Queue at the ticket booth
3. Age > 17 YO → condition
4. Could buy ticket
5. Could not buy ticket

Statement 4 will be processed if Statement 3 is met (Age > 17 YO)

Statement 5 will be processed if Statement 3 is not met (Age <= 17 YO)

→ Statements 4 and 5 are impossible to do both, only one will do.

SELECTION

Algorithm :

if condition

then

Action1 / Statement 1

else

Action2 / Statement 2

Action1/ Statement 1 will be processed if condition is met. If condition is not met, then Action2 / statement 2 will be processed.

REPETITION

Algorithm to print “Polman Astra” on the screen 10 times :

1. Print “Polman Astra” on the screen
2. Print “Polman Astra” on the screen
3. Print “Polman Astra” on the screen

…..

10. Print “Polman Astra” on the screen

Remember !! Algorithm should be simple.

Repeat 10 times:
    Print “Polman Astra” on the screen

REPETITION (*cont.*)

Algorithm :

Repeats the process for as many times

```
for i = 1 to 5
  Statement 1
  Statement 2
  Statement 3
```

Statement 1,2 and 3 will be repeated 5 times

```
for i = 1 to 10
  print “Polman Astra”
```

```
x = 1
b = 2
for i = 1 to 5
        a = b+2
        print a
        b=b+1
        x=x+1
```

Pseudo-Code Notation

A statement in descriptive notation:

```
write the value of x and y
```

pseudo-code notation:

```
Write (x, y)
```

C notation:

```
printf("%d %d", x, y);
```

A statement in descriptive notation:

```
input the value of x into y
```

pseudo-code notation:

```
y ← x
```

C notation:

```
y = x;
```

Algorithm Text Structure

```
PROGRAM Euclidean

{Given two non-negative integers m and n (m ≥ n). The Euclidean algorithm
looks for the greatest common divisor, GCD, of the two numbers, the largest
number that divides both of them without leaving a remainder.}

DECLARATION:
      m, n  : integer   { integers to be searched for their GCD }
      r     : integer   { remainder from division }

ALGORITHM:

Read (m, n)          { assume m ≥ n }
While n ≠ 0  do
      r  ← m mod n     { divide m with n, store the remainder in r }
      m ← n            { exchange m with n }
      n ← r            { exchange n with r }
End while
{ final condition after looping: n = 0 }

Write (m)
```

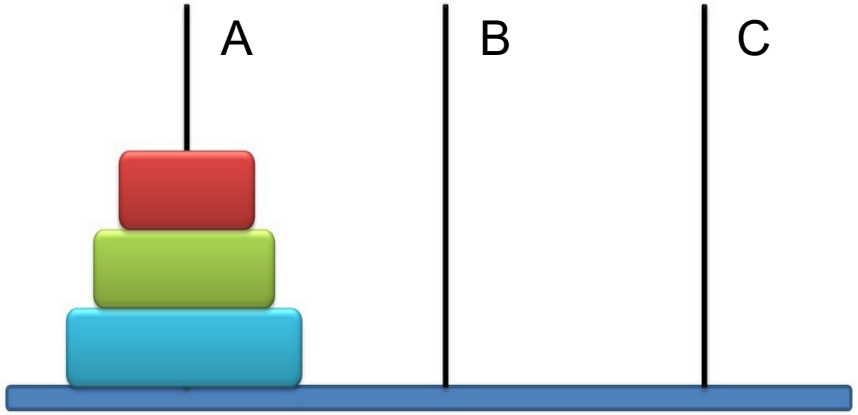
Title

Declaration

Algorithm

Practice

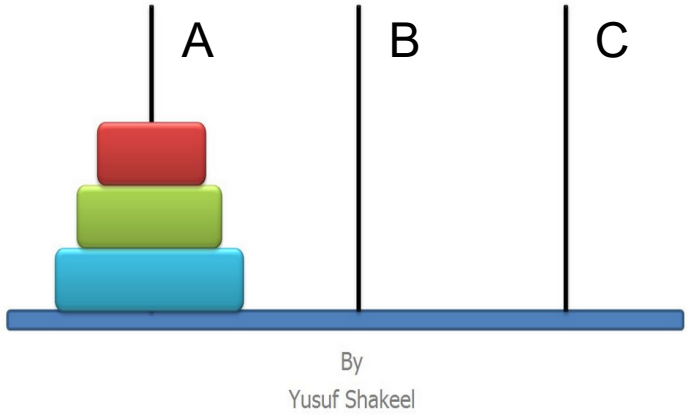
Tower of Hanoi Algorithm



Three discs each having different diameters have holes in their center point. The three discs are inserted in an A iron rod so that larger diameter discs are always located under smaller diameter disks. Write the algorithm in the descriptive notation to move the entire disk to the B iron rod. Each time only one disc is allowed to move, but at each move there should be no larger discs over the small disc. The C iron rod can be used as a transitional point while maintaining the rules already mentioned.

Practice

Tower of Hanoi Algorithm





Source: Wikipedia

Three discs each having different diameters have holes in their center point. The three discs are inserted in an A iron rod so that larger diameter discs are always located under smaller diameter disks. Write the algorithm in the descriptive notation to move the entire disk to the B iron rod. Each time only one disc is allowed to move, but at each move there should be no larger discs over the small disc. The C iron rod can be used as a transitional point while maintaining the rules already mentioned.

Bibliography

- Munir, Lidya, “Algoritma & Pemrograman”, 6th Ed, 2016 Penerbit Informatika.
- Kernighan, Ritchie, “The C Programming Language”, 2nd Ed, Prentice Hall.